Ming Li

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About Me

I am a Ph.D. candidate at the University of Central Florida and an intern at ByteDance-Seed (USA), working on reinforcement learning (RL) for multimodal generation, advised by Prof. Chen Chen.

During my internship, I contributed to RL post-training for the **Seedance** series (1.5, 2.0, 2.5), Seedream 5.0 Pro, and Seeworld (streaming interactive video world model).

Experience

Intern	ByteDance-Seed, USA	Mentors: *Jie Wu*, *Rui Wang*	01/2025 – Present
Intern	ByteDance Inc, USA	Mentor: *Sijie Zhu*	05/2024 – 12/2024
Intern	ByteDance Inc, China	Mentor: *Jie Wu*	01/2022 – 07/2023

Education

Ph.D.	Computer Science	University of Central Florida, USA	08/2023 – 12/2026
M.S.	Computer Science	Xiamen University, China	09/2020 – 06/2023
B.S.	Software Engineering	Hainan University, China	09/2016 – 06/2020

Industry Projects

Contributions to Seedance and Seedream below were shipped in the released models.

- **Seeworld** (work in progress) 07/2026 – Present
  Exploring how to design rewards and RL training for a streaming interactive video world model, where videos are generated autoregressively and new references can arrive at any time.
- **Seedance 2.5** [Project] 03/2026 – 06/2026
  - **Reward modeling:** Built the video-extension reward model pipeline, with training data built from scratch. Designed a frame sampling strategy for video extension, where the default uniform sampling gives poor reward accuracy. RL with this reward model improved video-extension quality in the final model.
  - **Agentic-Rubric RL:** Built a pipeline where an agent iteratively refines the rubrics so that a fixed verifier's judgments better match human annotations, and used these judgments as RL rewards. The consistency and overall performance in reference-to-video generation improved over the baseline model.
- **Seedance 2.0** [Project | Paper] 12/2025 – 02/2026
  - **DPO implementation and data:** Implemented DPO on the SFT training codebase for reference-to-video generation and built multi-dimensional preference data, improving content preservation, reference adherence, and overall quality over the ReFL-tuned model by up to +23% (win − loss rate) in human evaluation.
  - **Training recipe:** Conducted experiments on pair selection and data mixture to find what kind of preference data helps DPO training. Improved the vanilla DPO algorithm to keep it from only optimizing the preference margin while both chosen and rejected samples get worse.
- **Seedream 5.0 Pro** [Project] 04/2026 – 06/2026
  Applied Agentic-Rubric RL with multimodal context in the rubric to improve image vividness. In internal human evaluation, aesthetics went from on par with Nano-Banana-2 before RL to clearly ahead after RL.
- **RewardDance: Reward Scaling in Visual Generation** [Paper] 01/2025 – 09/2025
  Trained and scaled visual reward models from 1B to 26B parameters for image and video generation. Larger reward models mitigated reward hacking and improved downstream generation quality through ReFL.

RESEARCH PROJECTS

- **RL for Multimodal Generation:** ViPO (ICLR'26), ControlNet++ (ECCV'24)
  For controllable generation, where outputs often deviate from the conditional controls, proposed **ControlNet++**, which optimizes pixel-level cycle consistency with an efficient reward strategy, improving over ControlNet by 11.1% mIoU and 13.4% SSIM. For preference optimization, where DPO fails to scale on noisy data, proposed Poly-DPO in **ViPO**, which scales on such data (+6.87 GenEval over Diffusion-DPO).
- **Long-Video Generation:** Self-Forcing++ (ICLR'26), LoL (CVPR'26)
  To mitigate quality degradation beyond the teacher's horizon, **Self-Forcing++** lets the teacher guide segments sampled from self-generated long videos, reaching 4 minutes. To eliminate sink-collapse, where content reverts to the sink frame, **LoL** adds training-free multi-head RoPE jitter, enabling 12-hour streaming.
- **Image Editing:** SuperEdit (ICCV'25), Multi-Reward (ICLR'25)
  To resolve the instruction-image mismatch in auto-constructed editing data, proposed **SuperEdit**, which rectifies the instructions and adds contrastive ones, outperforming SmartEdit by 9.19% with 30× less data and a 13× smaller model. **Multi-Reward** further conditions the editing model on multi-perspective rewards.

SELECTED PUBLICATIONS (COMPLETE LIST...)

Total: 1100+ citations, 1000+ GitHub stars.

- **Seedance 2.0: Advancing Video Generation for World Complexity**
  Team Seedance (**Ming Li** as core contributor, RL)
  [ByteDance-Seed Tech Report | Project | Paper]
- **Seedance 1.5 Pro: A Native Audio-Visual Joint Generation Foundation Model**
  Team Seedance (**Ming Li** as core contributor, RL)
  [ByteDance-Seed Tech Report | Project | Paper]
- **RewardDance: Reward Scaling in Visual Generation**
  Jie Wu, Yu Gao, Zilyu Ye, **Ming Li**, Liang Li, Hanzhong Guo, Jie Liu, Zeyue Xue, Xiaoxia Hou, Wei Liu, Yan Zeng, Weilin Huang
  [ByteDance-Seed Tech Report | Paper]
- **ViPO: Visual Preference Optimization at Scale**
  Ming Li, Jie Wu, Justin Cui, Xiaojie Li, Rui Wang, Chen Chen
  [ICLR'26 | Project | Paper]
- **Self-Forcing++: Towards Minute-Scale High-Quality Video Generation**
  Justin Cui, Jie Wu, **Ming Li**, Tao Yang, Xiaojie Li, Rui Wang, Andrew Bai, Yuanhao Ban, Cho-Jui Hsieh
  [ICLR'26 | Project | Paper]
- **SuperEdit: Rectifying and Facilitating Supervision for Instruction-based Image Editing**
  Ming Li, Xin Gu, Fan Chen, Xiaoying Xing, Longyin Wen, Chen Chen, Sijie Zhu
  [ICCV'25 | Paper | Code | GitHub Stars 100+]
- **ControlNet++: Improving Conditional Controls with Efficient Consistency Feedback**
  Ming Li, Taojiannan Yang, Huafeng Kuang, Jie Wu, Zhaoning Wang, Xuefeng Xiao, Chen Chen
  [ECCV'24 | Project | Paper | Code | GitHub Stars 500+]
- **AlignDet: Aligning Pre-training and Fine-tuning in Object Detection**
  Ming Li, Jie Wu, Xionghui Wang, Chen Chen, Jie Qin, Xuefeng Xiao, Rui Wang, Min Zheng, Xin Pan
  [ICCV'23 | Project | Paper | Code | GitHub Stars 100+]

AWARDS AND HONORS

- Graduate Artificial Intelligence Initiative (Aii) Doctoral Fellowship, the University of Central Florida. 2023
- ORCGS Doctoral Fellowship, the University of Central Florida. 2023
- 1st Place, CVPR 2023 Long-form Video Understanding and Generation Challenge. 2023
- 1st Place, CVPR 2022 Accessibility Vision and Autonomy Challenge. 2022